Slot 14 board script: the table as a map of configurations

Clock: Launcher 0:00-6:00, Story and goal 6:00-9:00, Presentation 9:00-19:00, Guidance 19:00-26:00, Practice 26:00-41:00, Feedback 41:00-45:00, Exit slip 45:00-48:00, Up Next 48:00-50:00. Excalidraw stencils to open: the blank table, the energy ladder.

Review, sheet L1 to L4 (on the board as they arrive)

L1. Write the full configuration of lithium (Z = 3). L2. Write the full configuration of sodium (Z = 11). L3. Lithium and sodium are in the same group. What do the last pieces of their configurations have in common? L4. Guess, no penalty: how many elements are in the fourth row of the periodic table?

SAY 0:00–6:00
They work alone, notes face down. Answers: L1 $1s^2 2s^1$, L2 $1s^2 2s^2 2p^6 3s^1$, L3 both end in s^1: one electron in the outer shell (the slot 13 pattern, group 1). Take two answers aloud.
L4 is a guess, no penalty. Do not answer it. Return to it at 13:00 (live C1): the answer is 18.
APPLY: this is where the slot 10 shorthand ([He]$2s^1$, [Ne]$3s^1$) pays off: the ends match.

The table was a map before anyone could read it

In 1869 Mendeleev set the elements in rows by mass and by how they behaved, and left gaps for elements nobody had found. It worked, and nobody knew why. Around 1913 Moseley showed that the true order was by atomic number. Within about fifteen years the shell model gave the reason: the rows are shells filling and the columns are the electrons left over in the outer shell. Today we draw the table from the configurations we wrote in slots 9 and 10, and then read it backwards.

SAY 6:00–9:00
Ninety seconds, no more. Mendeleev placed elements without a reason; we can say why. Hedge the dates: "around 1913," "within about fifteen years."
EXPERT: Mendeleev ordered by mass and swapped Te and I to keep the chemistry right. The atomic number explained the swaps.

Big question. If I tell you only where an element sits on the table, what can you say about its electrons?

SAY 8:00
Write the big question on the board, in the box. Return to it at the keepers and at the end.

A. Build the table from the ladder

Fill the subshells in the order from slot 9: 1*s*, 2*s*, 2*p*, 3*s*, 3*p*, 4*s*, 3*d*, 4*p*. Start a new row each time the outer shell number goes up. That gives four rows.

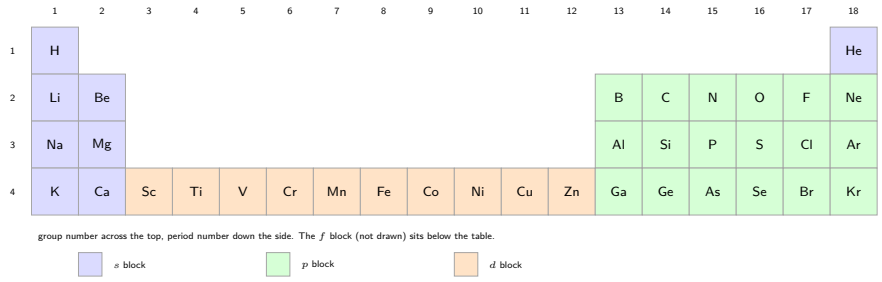
Period (outer shell)	Subshells that fill, in order	Elements
1	1*s*	2
2	2*s* 2*p*	8
3	3*s* 3*p*	8
4	4*s* 3*d* 4*p*	18

Period A row. Its number is the outer shell number, so period 3 means the outer electrons are in shell 3.

Block The subshell that fills last. The *s* block is groups 1 and 2. The *p* block is groups 13 to 18. The *d* block is groups 3 to 12.

The *d* subshell is one shell behind its period. The 3*d* boxes fill after 4*s*, so they sit in period 4.

SAY 9:00–14:00
Slow. Fill the ladder on the board in the slot 9 order and put the cut after 1s, after 2p, after 3p, after 4p. Row sizes come out 2, 8, 8, 18 by counting boxes. Do not tell them the rows; let them count. EXCALIDRAW: use the energy-ladder stencil, then the blank-table stencil.
COPY 1, COPY 2, COPY 3
ASK 13:00 (whiteboards, sample two): sheet C1. Period 3 has 8 elements and period 4 has 18. What does period 4 have that period 3 does not? Check the L4 guess. Answer: the 3d subshell, ten more boxes.
WATCH FOR: "Period 4 has 8 elements like the others." Ask: where did the 3d electrons go?
WATCH FOR: "The d block starts at shell 3." 3d is in period 4. Point at the ladder.
EXPERT: the 4s-before-3d order is a good approximation, not a law. Cr and Cu break it. Not taught; do not volunteer.



SAY
The table through krypton ($Z = 36$). Students sketch a copy in the notebook, symbols for the first two rows only, blocks shaded. The full table with symbols is on the sheet.

B. Valence configuration from position

Method. 1. Find the period n and the group g.
2. Name the block: groups 1 and 2 are s, groups 13 to 18 are p.
3. Group 1 gives ns^1. Group 2 gives ns^2. Groups 13 to 18 give ns^2np^{g-12}.
4. Valence electrons: g for groups 1 and 2, and $g - 10$ for groups 13 to 18.

One exception. Helium is in group 18 but it is $1s^2$, with only 2 valence electrons, because there is no $1p$ subshell.

SAY 14:00–19:00
Mr. D derives it on the board once, with Cl: period 3, group 17, so $3s^23p^5$ and 7 valence. Then the pattern for every group.
COPY 4, COPY 5, COPY 6
ASK 17:00 (whiteboards, sample two): sheet C2. Which block are potassium, iron and gallium in? Answer: s, d, p.
COPY 7
WATCH FOR: "Helium has 8 valence electrons, it is a noble gas." Ask: how many boxes does shell 1 have?
ASK 18:30 (whiteboards): sheet C3. Chlorine is in group 17. Write its valence configuration and count its valence electrons. Answer: $3s^23p^5$, 7. This is the last live question; everything else is on the sheet.

Worked together

Example (Sulfur, together)**.** Sulfur is in period 3, group 16. Group 16 is in the p block, so the valence configuration is $ns^2np^{g-12} = 3s^23p^4$. Valence electrons: $16 - 10 = 6$.

Example (Gallium, first step done)**.** Gallium is in period 4, group 13. It is in the p block. Finish it: $4s^24p^1$, with $13 - 10 = 3$ valence electrons.

SAY 19:00–26:00
Sulfur with the class, every step spoken. Then gallium: the first step (the p block) is on the board; students finish alone on whiteboards, then compare with a neighbor.
COPY 8, COPY 9
APPLY: both answers are on the board for the whole practice block so students can check themselves.

What to copy

Copy 1. Period = row = the outer shell number. Rows hold 2, 8, 8, 18.

Copy 2. Block = the subshell filling last: s is groups 1–2, p is groups 13–18, d is groups 3–12.

Copy 3. The d subshell is one shell behind its period: $3d$ sits in period 4.

Copy 4. Group 1: ns^1. Group 2: ns^2.

Copy 5. Groups 13–18: ns^2np^{g-12}, where g is the group number.

Copy 6. Valence electrons: groups 1–2 equal g; groups 13–18 equal $g - 10$.

Copy 7. Helium is in group 18 but is $1s^2$: 2 valence electrons.

Copy 8. Sulfur: period 3, group 16 gives $3s^23p^4$ (6 valence).

Copy 9. Gallium: period 4, group 13 gives $4s^24p^1$ (3 valence).

COPY
Nine keepers, spoken as they come up. Everything else is spoken. Notes face down for the practice sheet.
SHEET 26:00–41:00: the sheet has four modules: Review L1 to L4 (done), Concept C1 to C6 (C1 to C3 were live), Practice P1 to P9, Challenge H1 to H4 (spice 3 and 4). Core is C4 to C6 and P1 to P5 (about 14 minutes); P6 to P9 is the bank; Challenge is for finishers. Answers in the notebook, numbered, one under another. On a 3 or 4 chili problem they write what they tried before you give a nudge (key has the hint ladders).
WATCH FOR (sheet P5): students stop at 3d and call nickel period 3. ASK: what is the highest shell number?
SIM: if the room is behind, swap Gallium for Aluminum in guidance (period 3, group 13: $3s^23p^1$).

Feedback, 41:00–45:00. Go through C4, P2 and P5 on the board; then the one error most common on the sheet. **Exit slip, 45:00–48:00.** Handed out face down; students draw on paper.

SAY 41:00–45:00
Answers: the key. Spend the time on the common block error (call a d-block atom a p-block atom, or the other way).
EXIT 45:00–48:00: a drawing. They sketch the table frame (they know it now), mark the position, shade the block, and write the valence configuration. Scale on the key.

Up next

Lithium and sodium both end in s^1, and both react the same way. But sodium gives up its outer electron more easily. Why is sodium's outer electron easier to remove than lithium's?

UP NEXT 48:00–50:00
Ask it, do not answer it. Slot 15 is the sim that answers it. Collect the exit slips at the door.